Assignment 11

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Given that X and Y are independent, $E(X) = E(Y) = \mu$, $\mathrm{Var}(X) = \sigma_X^2$ and $\mathrm{Var}(Y) = \sigma_Y^2$.

a. We have

$$\begin{aligned} E(Z) &= E(\alpha X + (1-\alpha)Y) \\ &= \alpha E(X) + (1-\alpha)E(Y) \\ &= \alpha\mu + (1-\alpha)\mu \\ &= \mu \end{aligned}$$

b. Since X and Y are independent,

$$\begin{aligned} \mathrm{Var}(Z) &= \mathrm{Var}(\alpha X + (1-\alpha)Y) \\ &= \alpha^2\mathrm{Var}(X) + (1-\alpha)^2\mathrm{Var}(Y) \\ &= \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 \end{aligned}$$

To minimize $\mathrm{Var}(Z)$, we take the derivative with respect to α and set it to 0:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\alpha}\mathrm{Var}(Z) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 &= 0 \\ \alpha\sigma_X^2 &= (1-\alpha)\sigma_Y^2 \\ \alpha(\sigma_X^2 + \sigma_Y^2) &= \sigma_Y^2 \\ \alpha &= \frac{\sigma_Y^2}{\sigma_X^2 + \sigma_Y^2} \end{aligned}$$

The second derivative is $2\sigma_X^2 + 2\sigma_Y^2 > 0$, so this α indeed minimizes the variance.

c. For the simple average, we set $\alpha = 1/2$. The variance of the simple average is

$$\mathrm{Var}\left(\frac{X+Y}{2}\right) = \frac{\sigma_X^2 + \sigma_Y^2}{4}.$$

To make it better (i.e. having a smaller variance) than using either X or Y alone, we need:

$$\begin{cases} \frac{\sigma_X^2+\sigma_Y^2}{4} < \sigma_X^2 \\ \frac{\sigma_X^2+\sigma_Y^2}{4} < \sigma_Y^2 \end{cases}$$

This simplifies to:

$$\begin{cases} \sigma_Y^2 < 3\sigma_X^2 \\ \sigma_X^2 < 3\sigma_Y^2 \end{cases} \implies \frac{1}{3}\sigma_X^2 < \sigma_Y^2 < 3\sigma_X^2.$$

So the condition is $\frac{1}{3}\sigma_X^2 < \sigma_Y^2 < 3\sigma_X^2$.

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$$E(\bar{X}) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \mu$$

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{\sigma^2}{n}$$

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$$\begin{aligned} E(T) &= E\left(\sum_{k=1}^{n} kX_k\right) \\ &= \sum_{k=1}^{n} kE(X_k) \\ &= \mu\sum_{k=1}^{n} k \\ &= \frac{n(n+1)}{2}\mu \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(T) &= \mathrm{Var}\left(\sum_{k=1}^{n} kX_k\right) \\ &= \sum_{k=1}^{n} k^2\mathrm{Var}(X_k) \\ &= \sigma^2\sum_{k=1}^{n} k^2 \\ &= \sigma^2\frac{n(n+1)(2n+1)}{6} \end{aligned}$$

4 Supplementary Question 1.1

$$E(Z) = E(5X - 2Y + 15) = 5E(X) - 2E(Y) + 15 = 5 \times 3 - 2 \times 1 + 15 = 28$$

$$D(Z) = D(5X - 2Y + 15) = 25D(X) + 4D(Y) = 25 \times 4 + 4 \times 9 = 100 + 36 = 136$$

5 Supplementary Question 1.2

Given that $E(X_i) = 2i$, $D(X_i) = 5 - i$, and X_i are independent for $i = 1, 2, 3, 4$, we have

$$E(Z) = E(2X_1 - X_2 + 3X_3 - 0.5X_4) = 2E(X_1) - E(X_2) + 3E(X_3) - 0.5E(X_4) = 2\times2 - 4 + 3\times6 - 0.5\times8 = 14$$

$$\begin{aligned} D(Z) &= D(2X_1 - X_2 + 3X_3 - 0.5X_4) \\ &= 4D(X_1) + D(X_2) + 9D(X_3) + 0.25D(X_4) \\ &= 4 \times 4 + 3 + 9 \times 2 + 0.25 \times 1 \\ &= 16 + 3 + 18 + 0.25 = 37.25 \end{aligned}$$

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Given that $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$, $Z \sim N(\mu_Z, \sigma_Z^2)$, where they are uncorrelated, for

$$\begin{aligned} U &= Z + X \\ V &= Z + Y \end{aligned}$$

we have

$$\begin{aligned} \mathrm{Cov}(U, V) &= \mathrm{Cov}(Z + X, Z + Y) \\ &= \mathrm{Cov}(Z, Z) + \mathrm{Cov}(Z, Y) + \mathrm{Cov}(X, Z) + \mathrm{Cov}(X, Y) \\ &= \sigma_Z^2 \end{aligned}$$

and

$$\begin{aligned} \rho_{UV} &= \frac{\mathrm{Cov}(U, V)}{\sqrt{\mathrm{Var}(U)\mathrm{Var}(V)}} \\ &= \frac{\sigma_Z^2}{\sqrt{(\sigma_X^2 + \sigma_Z^2)(\sigma_Y^2 + \sigma_Z^2)}} \end{aligned}$$

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Given that the density function have the symmetric property $f(y) = f(-y)$ then $E(Y) = 0$ and $X = SY$, where S is a random variable independent variable taking on the value $+1$ and -1 with probability $\frac{1}{2}$, we have

$$\begin{aligned} \mathrm{Cov}(X, Y) &= E(XY) - E(X)E(Y) \\ &= E(SY^2) - E(S)E(Y)E(Y) \\ &= E(S)E(Y^2) - E(S)E(Y)E(Y) \\ &= 0 \end{aligned}$$

We show that X and Y are not independent. Since $X = SY$,

$$\begin{aligned} P(X = Y) &= P(S = 1) + P(S = -1, Y = 0) \\ &= \tfrac{1}{2} + 0 = \tfrac{1}{2}, \end{aligned}$$

because Y has a density and thus $P(Y = 0) = 0$. If X and Y were independent and each had a density, then $P(X = Y) = 0$, a contradiction. Hence X and Y are not independent.

8 Extra Question

8.1 1)

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} \frac{x}{2} e^{-|x|} \mathrm{d}x \\ &= \int_{0}^{\infty} \frac{x}{2} e^{-x} \mathrm{d}x + \int_{-\infty}^{0} \frac{x}{2} e^{x} \mathrm{d}x \\ &= 0 \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(X) &= E(X^2) - E(X)^2 \\ &= \int_{-\infty}^{\infty} \frac{x^2}{2} e^{-|x|} \mathrm{d}x \\ &= \int_{0}^{\infty} \frac{x^2}{2} e^{-x} \mathrm{d}x + \int_{-\infty}^{0} \frac{x^2}{2} e^{x} \mathrm{d}x \\ &= 2 \end{aligned}$$

8.2 2)

No, X and $|X|$ are not independent. If we know the value of $|X|$, the value of X is completely restricted to $\pm|X|$. For instance,

$$P(X \geq 1 \mid |X| < 1) = 0$$

but $P(X \geq 1) > 0$. Since the conditional probability is not equal to the marginal probability, they are dependent.

8.3 3)

They are uncorrelated. To check for correlation, we compute the covariance:

$$\mathrm{Cov}(X, |X|) = E(X|X|) - E(X)E(|X|)$$

We already computed that $E(X) = 0$. Furthermore,

$$E(X|X|) = \int_{-\infty}^{\infty} x|x|\frac{1}{2}e^{-|x|}\mathrm{d}x$$

Notice that $g(x) = x|x|$ is an odd function, and the density $f(x) = \frac{1}{2}e^{-|x|}$ is an even function. Their product is an odd function. Integrating an odd function over the symmetric interval $(-\infty, \infty)$ gives 0. Hence, $E(X|X|) = 0$.

Therefore,

$$\mathrm{Cov}(X, |X|) = 0 - 0 \cdot E(|X|) = 0$$

Since the covariance is zero, X and $|X|$ are uncorrelated (not correlated).

9 Supplementary Question 2.1

$$f_X(x) = \int_0^2 \frac{x+y}{8}\mathrm{d}y = \frac{x}{4} + \frac{1}{4}$$

$$f_Y(y) = \int_0^2 \frac{x+y}{8}\mathrm{d}x = \frac{y}{4} + \frac{1}{4}$$

$$E(X) = \int_0^2 x\left(\frac{x}{4} + \frac{1}{4}\right)\mathrm{d}x = \frac{7}{6}$$

$$E(Y) = \int_0^2 y\left(\frac{y}{4} + \frac{1}{4}\right)\mathrm{d}y = \frac{7}{6}$$

$$\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y) = \int_0^2\int_0^2 xy\frac{x+y}{8}\mathrm{d}x\mathrm{d}y - \frac{7}{6}\cdot\frac{7}{6} = \frac{4}{3} - \frac{49}{36} = -\frac{1}{36}$$

$$\begin{aligned}
\rho_{XY} &= \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} \\
&= \frac{-\frac{1}{36}}{\sqrt{\left(\int_0^2 x^2\left(\frac{x}{4}+\frac{1}{4}\right)\mathrm{d}x - \left(\frac{7}{6}\right)^2\right)\left(\int_0^2 y^2\left(\frac{y}{4}+\frac{1}{4}\right)\mathrm{d}y - \left(\frac{7}{6}\right)^2\right)}} \\
&= \frac{-\frac{1}{36}}{\sqrt{\left(\frac{5}{3} - \frac{49}{36}\right)^2}} \\
&= -\frac{1}{11}
\end{aligned}$$

$$\begin{aligned}
D(X+Y) &= D(X) + D(Y) + 2\mathrm{Cov}(X,Y) \\
&= \frac{11}{36} + \frac{11}{36} + 2\cdot\left(-\frac{1}{36}\right) \\
&= \frac{5}{9}
\end{aligned}$$

10 Supplementary Question 2.2

Given that $X, Y \sim N(\mu, sigma)$ and X, Y are independent. Let

$$Z = \alpha X + \beta Y, W = \alpha X - \beta Y$$

then

$$\begin{aligned}
\text{Cov}(Z, W) &= E(ZW) - E(Z)E(W) \\
&= E((\alpha X + \beta Y)(\alpha X - \beta Y)) - E(\alpha X + \beta Y)E(\alpha X - \beta Y) \\
&= E(\alpha^2 X^2 - \beta^2 Y^2) - E(\alpha X + \beta Y)E(\alpha X - \beta Y) \\
&= \alpha^2 E(X^2) - \beta^2 E(Y^2) - (\alpha E(X) + \beta E(Y))(\alpha E(X) - \beta E(Y)) \\
&= \alpha^2(\sigma^2 + \mu^2) - \beta^2(\sigma^2 + \mu^2) - (\alpha\mu + \beta\mu)(\alpha\mu - \beta\mu) \\
&= (\alpha^2 - \beta^2)(\sigma^2 + \mu^2) - (\alpha^2 - \beta^2)\mu^2 \\
&= (\alpha^2 - \beta^2)\sigma^2
\end{aligned}$$

$$\begin{aligned}
\rho_{ZW} &= \frac{\text{Cov}(Z, W)}{\sqrt{\text{Var}(Z)\text{Var}(W)}} \\
&= \frac{(\alpha^2 - \beta^2)\sigma^2}{\sqrt{(\alpha^2\sigma^2 + \beta^2\sigma^2)(\alpha^2\sigma^2 + \beta^2\sigma^2)}} \\
&= \frac{(\alpha^2 - \beta^2)\sigma^2}{\sqrt{(\alpha^2 + \beta^2)^2\sigma^4}} \\
&= \frac{\alpha^2 - \beta^2}{\alpha^2 + \beta^2}
\end{aligned}$$